

Pitfalls and Hurdles

Standard Pitfalls

- ☐ Solving the World
 - ☐ Manna from Heaven
 - ☐ Computer Bum
 - ☐ Yet Another Language
 - ☐ Micawberism
 - ☐ Ivory Tower
 - ☐ Misunderstood Genius
 - ☐ Lost in Abstraction
 - ☐ Ambition Paralysis
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Solving the World

- ❑ Most students pick research goals which are far too ambitious.
 - ❑ You should read the literature and talk to fellow workers to find out what the state of the art is.
 - ❑ One good source of ideas is the further work sections of papers. Read the literature critically.
 - ❑ Another good source is redoing bad work, which contained the germ of a good idea.
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Manna from Heaven

- Having chosen a topic, what do you do next? It is no good sitting in your room with a blank piece of paper and a pencil, expecting good ideas to come from above.

 - What you can do is:
 - (a) Complete an initial electronic literature search in your chosen topic/field.
 - (b) Use electronic sources - papers published on the WWW and through subscriptions to relevant News Groups.
 - (c) Read the literature. Read with a question in mind, e.g. What is wrong with this? How can I use this? etc
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Manna from Heaven

- (d) Talk to people. Do not go away and hide. Do not be ashamed of your ideas. Other people's are sillier.
 - (e) Tackle a simplified version of your problem. Ask your supervisor for exercises, mini-projects, etc.
 - (f) Write down your ideas in a working paper. Imagine yourself explaining your ideas to someone.
 - (g) Give a seminar to your research group.
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- You will be amazed at how half-baked ideas take shape and bugs are exposed.
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Computer Bum

- ❑ Computers are very seductive - spend years at a terminal debugging your programs and tuning up the input/output routines; get a satisfying sense of achievement when a bug is exposed or a nice output generated.
 - ❑ This is illusory! Your work must be explainable at a higher level than code, for it to make a real contribution to knowledge.
 - ❑ Try to plan your program theoretically before going to the computer. Undertake some of Einstein's 'thought experiments'. If you must work some of it out at the keyboard then rush away soon and work out the theory.
 - ❑ Try to describe how it works to a friend; in a paper or at a seminar. If people do not understand it is your fault -- try harder.
 - ❑ Try to explain your ideas to a non-computer person. The necessary simplification will expose the framework of your ideas.
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Yet Another Language

- ❑ A terminal case of 'computer bum' is to get involved in writing yet another programming language.
 - ❑ Of course the existing languages do not offer exactly what you need for your project, but that is no excuse for writing another one.
 - ❑ You can usually find a reasonable candidate and add to it what you need. No one will use the language you write -- not even you!
 - ❑ You will have spent all your time on the language and none on the project you started with.
 - ❑ If you really believe that existing languages are inadequate then write a paper on it, carefully describing the deficiencies.
 - ❑ If you do a good enough demolition job, enhanced languages will spring up overnight.
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Micawberism

- ☐ Habitually optimistic point of view of a Micawber
 - ☐ ~~Gathering experimental data can be fun and gives all the appearance of productive work.~~
 - ☐ Make sure that you know what class of result you are attempting to establish with your experiments.
 - ☐ (a) Talk to people, explain what you think your experiment might show.
 - ☐ (b) Imagine the experiment finished and you have 'the data', what exactly would you do with it.
 - ☐ (c) Not only try out the experiment on one or two people first, but also try out the analysis.
 - ☐ Don't keep running experiments in the hope that something will turn up.
 - ☐ Use appropriate evaluation tool (may be third party) for establishment of effectiveness of your method
 - ☐ Such tools also enable to compare performance
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Ivory Tower

- Single-minded dedication to your topic is a good thing, but do not shut out the rest of the world completely.
 - Keep in touch with the state of the art in related fields.
 - Talk to other people about their research.
 - Attend selected seminars and lectures.
 - Set aside a part of the week for reading reviews and abstracts and skimming papers.
 - Working in a group is often found beneficial
 - Saves time & resource, improves quality
 - Instead of exploring all possible alternatives can save time by gaining experience from other team members
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Misunderstood Genius

- It is all too easy to believe that the reason why no one understands your ideas is because you are a genius and the others are all looneys and charlatans.
 - There are alternative causes for misunderstanding that you should consider:
 - Love of jargon.
 - Try to rephrase your ideas using ordinary English; try to rephrase your ideas in someone else's jargon. Do they come out any different?
 - If I can do it, it's trivial. Once you have seen the solution to a problem it appears trivial. Then it is tempting to say 'that's too easy, I'll try something else'. This is a nonterminating loop!
 - Motto: do the easiest thing first, then stand on shoulders and do the next easiest thing etc. -- a better infinite loop.
 - Use of appropriate examples
 - Use of appropriate scenarios
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Ambition Paralysis

- ❑ It is good to have high standards for your finished product
 - ❑ But do not apply the same standards to your initial attempts or you may never get started.
 - ❑ Do something simple, then apply your standards to refine it into something worthy.
 - ❑ Put your continuous effort to improve the quality of the final work
 - ❑ Believe in implementation and try to publish your work as supporting tool
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Methodology ?

- Computer Science is relatively young field, and is somewhat interdisciplinary, it does not have one received framework or one well defined methodology.
 - One of the (difficult) tasks that you face as a research student is the development, consciously or unconsciously, of a suitable approach to the problem(s) being tackled.
 - In the course of evolving an appropriate methodology, you will encounter many other methodologies and philosophical positions, many of which will seem outrageous or hopelessly misguided.
 - You will nevertheless find that these bizarre viewpoints have strong proponents, perhaps at the next desk in your lab!
 - Hence, much of the formative period for your own methodology is spent having violent arguments with fellow researchers who are promoting alternative views.
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Methodology ?

- Out of this struggle, your reading, your attendance at seminars, your debugging, and other hard work, will emerge your world view on your area and related philosophical issues.
 - In later years, you will probably come to take this outlook for granted, perhaps modifying it occasionally in some way.
 - However, it is quite likely to loom very large in your life during the period of your project, and when you come to write your thesis you may feel compelled to expand upon your philosophy of life at length.
 - Restrain yourself the examiners won't be all that interested.
 - Give a brief summary of your methodological assumptions, giving references across to existing arguments or frameworks where appropriate.
 - Confine yourself to the points which are essential to the understanding of the substance of your thesis.
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Discovery Route?

- In the course of your project, you will come to certain beliefs about technical issues, some of which will be novel, and many of which will be rediscoveries (or new understandings) of established concepts.
 - In presenting your thesis, it is important to distinguish between the justification (for instance, generality, efficiency, perspicuity, practicality) for some position or technique, and the route by which you happened to come to favour this idea.
 - The readers and examiners aren't particularly interested in reconstructing how you became convinced of an idea. They are interested in the general arguments in favour of the idea.
 - When you have just become convinced of some point, your own discovery route will seem like the dominant reason for it.
 - You may need a cooling off period before you can detach yourself sufficiently to write a reasoned support for the idea, particularly if it is your own idea as opposed to enthusiasm for someone else's.
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Sidetracks

- ❑ You will be tempted from time to time to take diversions from your core work because you feel that you have made a significant discovery that will change the world.
 - ❑ Beware of the sidetrack - it can consume months of fruitless distraction.
 - ❑ On the other hand, these sidetracks sometimes become the basis for significant new directions in your work.
 - ❑ Present these ideas to your supervisor or in a seminar as it might be fruitful.
 - ❑ Beware the sidetrack after the first 12 months and plan to pick it up when you have finished.
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The Development Trap

- ❑ Remember that research is not development, although development will be necessary in solving your problem.
 - ❑ If you stray too far into the big D, you will probably be passed by industrial R&D and find your years of work will be released as a product before you start to write-up.
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